

Jie Gao

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Malone Postdoctoral Fellow

Department of Computer Science & Malone Center for Engineering in Healthcare

Johns Hopkins University, Malone Hall, 3400 N Charles St, Baltimore, MD 21218, USA

Work Experience

- 2025.03–Present **Johns Hopkins University**, Baltimore, MD, USA
Malone Postdoctoral Fellow
Advisors: Mark Dredze, Ziang Xiao, Chien-Ming Huang
- 2024.02–2025.02 **Singapore-MIT Alliance for Research and Technology (SMART)**, Singapore
Postdoctoral Associate, Mens Manus Machina (M3S) team
Advisor: Thomas W. Malone

Education & Experience

- 2019.09–2024.01 **Singapore University of Technology and Design**, Singapore
Ph.D. in Information Systems Technology and Design
Advisor: Simon Perrault
- 2023.01–2023.07 **University of Notre Dame**, Indiana, USA
Visiting Ph.D. Student
Advisor: Toby Jia-jun Li
- 2022.06–2022.12 **National University of Singapore**, Singapore
Visiting Ph.D. Student
Advisor: Shengdong Zhao
- 2017.09–2019.06 **Zhejiang University**, Hangzhou, China
M.Eng. in Software Engineering (Information Product Design)
- 2013.09–2017.06 **Shanghai University**, Shanghai, China
B.Eng. in Digital Media Technology

Publications

- 2024 **CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models**
[Jie Gao](#), Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, Simon Tangi Perrault.
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)
- 2024 **Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms**
ShunYi Yeo, Gionnieve Lim, [Jie Gao](#), Weiyu Zhang, Simon Tangi Perrault.
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)
- 2023 **CoAICoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis**
[Jie Gao](#), Kenny Tsu Wei Choo, Junming Cao, Roy Ka Wei Lee, Simon Perrault.
ACM Transactions on Computer-Human Interaction (TOCHI 2023)
- 2023 **VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping**
Zheng Zhang, [Jie Gao](#), Ranjodh Singh Dhaliwal, Toby Jia-Jun Li.
Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2023)
- 2023 **GlassMessaging: Towards Ubiquitous Messaging Using OHMDs**
Nuwan Janaka, [Jie Gao](#), Lin Zhu, Shengdong Zhao, et al.
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2023)
- 2023 **Understanding the Complexity and Its Impact on Testing in ML-Enabled Systems**
Junming Cao, Bihuan Chen, Longjie Hu, [Jie Gao](#), et al.
IEEE International Conference on Software Maintenance and Evolution (ICSME 2023)

Late-Breaking Works, Workshops, Posters, Preprints

- 2024 **A Taxonomy for Human-LLM Interaction Modes: An Initial Exploration**
Jie Gao, Simret Araya Gebreegziabher, et al.
Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA 2024)
- 2024 **LLMs as Research Tools: Applications and Evaluations in HCI Data Work**
Marianne Aubin Le Quéré, Hope Schroeder, Casey Randazzo, Jie Gao, Ziv Epstein, Simon Perrault, David Mimno, Louise Barkhuus, Hanlin Li.
Workshop on LLMs as Research Tools at the ACM Conference on Human Factors in Computing Systems (CHI Workshop 2024)
- 2023 **CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis**
Jie Gao, Yuchen Guo, Toby Jia-jun Li, Simon Perrault.
ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW Demo 2023)
- 2022 **Differences of Challenges of Working from Home (WFH) between China and the US during COVID-19**
Jie Gao, Xiayin Ying, Junming Cao, et al.
Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA 2022)
- 2022 **From 996 to 007: Challenges of Working from Home During the Epidemic in China**
Jie Gao, Pin Sym Foong, Yifan Yang, et al.
arXiv preprint, arXiv:2201.09045
- 2018 **Expressive Plant: A Multisensory Interactive System for Sensory Training of Children with Autism**
Jie Gao, Leijing Zhou, Miaomiao Dong, Fan Zhang.
Poster presented at ACM International Joint Conference on Pervasive and Ubiquitous Computing (Ubicomp Poster 2018)

Talks and Presentations

- 2025.04 Designing Human-AI Teams to Extract Insights from Unstructured Data—Methods and Applications
HCI Seminar at Johns Hopkins University
- 2025.03 Human-Centric Software Engineering in the Era of Large Language Models
Singapore Management University
- 2025.02 Extracting Insights from (Un)structured Data: Methods and Applications
M3S Seminar in Mens Manus Machina (M3S) team at Singapore-MIT Alliance for Research and Technology

Academic Service

Reviewer, *Computers in Human Behavior Reports 2025*
Reviewer, *International Journal of Human-Computer Interaction (IJHCI2025)*
Associate Chair, *ACM Conference on Human Factors in Computing Systems (CHI 2025), Main Track*
Reviewer, *ACM Transactions on Interactive Intelligent Systems (TiiS 2024, 2025)*
Associate Chair, *ACM Conference on Human Factors in Computing Systems (CHI 2023), Late-Breaking Work Track*
Associate Chair, *ACM Conference on Human Factors in Computing Systems (CHI 2022), Late-Breaking Work Track*
Reviewer, *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2022, 2023)*
Reviewer, *ACM Conference on Human Factors in Computing Systems (CHI 2023, 2024)*
Reviewer, *International Conference on Artificial Intelligence in Education (AIED 2023)*

Mentoring Experience

- Summer-Spring
2024–2025 **Zhiyao Shu (Master, Remote Research Intern)**
University of California, Berkeley (Graduated). Now incoming PhD student at George Mason University
- Summer–Winter
2024 **Xiaoxian Zhang (Master, Research Intern)**
National University of Singapore

Summer 2024	Erika Lee (Undergraduate, Research Intern) University of California, San Diego
Fall 2022–Spring 2023	Yuchen Guo (Undergraduate, Research Intern) Singapore University of Technology and Design (Graduated)
Fall 2022–Spring 2023	Tianqin Zhang (Undergraduate, Research Intern) Singapore University of Technology and Design

Teaching Experience

Fall 2020	TA, 50.002: Introduction to Algorithms (Singapore University of Technology and Design)
Summer 2020	TA, 50.017: Graphics and Visualization (Singapore University of Technology and Design)

Technical Skills

Programming Languages & Development

Python, JavaScript, HTML/CSS, $\LaTeX$, React.js, Node.js, MongoDB, Git, etc.

Data Analysis & Machine Learning

Pandas, Numpy, SciPy, scikit-learn, PyTorch, NLTK, Matplotlib, Seaborn

Qualitative Analysis

Grounded Theory, Thematic Analysis, Atlas.ti, etc.

Tools & Platforms

Photoshop, Illustrator, Premiere, etc.

Languages

English (fluent), Mandarin (native)